Misr University for Science and Technology
College of Nursing
Medical Surgical Nursing Department

Clinical Practice Medical Surgical II
NUR 204 P

Contributors

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Procedure
1. Wound care.
2. Surgical suture removal.
3. IV therapy
4. Chest physiotherapy
5. NGT insertion& care.

Applying a Dry Sterile Dressing

Equipment:

- Clean disposable gloves
- Sterile gloves
- Moisture-proof bag
- Bath blanket
- Tape
- Adhesive remover or cotton with saline
- Cleansing solution (premedical order or agency protocol)
- Sterile dressing set(if not available ,gather the following:
  - ✓ Sterile drape
  - ✓ Gauze dressing(Surgi pads)
  - ✓ Sterile container for cleansing solution
  - ✓ Precut gauze if a drain is present

Procedure:

Action	Rationale
1. Review medical orders for dressing change procedure and list of needed supplies.	A physician's order is needed to prescribe a cleansing solution and individualized patient needs, such as type and amount of dressing Supplies needed.
2.Explaintheproceduretothepatient.	Patient cooperation is necessary to avoid contamination of the wound. Explanation decreases anxiety and increases cooperation.
3- Gather the equipment	-Organization. Promotes efficient time management
4- wash your hand	Hand washing decrease the spread of microorganisms.
5- keep patients privacy ;-Close the door or curtain when exposing the area	[-Closing the door or curtain provides privacy and warmth.
6. Position the patient. Using a bath blanket, drape the patient so that only the wound is exposed.	Provides privacy and prevents chilling.
7. Place the moisture-proof bag within easy reach. Make a cuff on the bag by folding the top over.	Provides a receptacle for proper disposal of contaminated dressings.
8.Assist the patient to a comfortable position that provides easy access to the wound area.	-Consideration for the comfort of the client is primary concern. The nurse promotes good body mechanics to avoid muscle strain.
9.Weardisposablegloves and Loosen the tape on the dressing use adhesive removal if necessary.	Prevents transmission of microorganisms. -It is easier to loosen tape before putting on gloves.

a. If Montgomery straps or a binder were used, untie the tapes. If tape was used, gently remove tape by pulling up small sections at a time while holding down the skin in front of the tape (provides countertraction on the skin). If resistance is met, you may need to use adhesive remover (if skin is torn during tape removal, you have created another wound). **b.** Carefully remove the outer protective dressing. Then remove the inner layers of gauze. If there is a drain present, use caution so that drains are not accidentally removed or dislodged. **c.** Place the soiled dressings and disposable gloves in the moisture-proof bag.	Careful removal of adhesive tapes prevents skin breakdown. Exposes the wound. Provides proper disposal and prevents contamination.
10. Assessthewound; note the odor, presence of any drainage, amount, color and consistency	The wound needs to be assessed For signs of complications and healing.
11. Opensteriledressingtrayorsetupsterile supplies and cleansing solution.	Maintains sterile technique.
12. Poursolutionintosterilebasin.	Keeps supplies sterile.
13. Donsterile gloves.	Sterile gloves are needed if the wound is open and if drains are present to prevent introduction of microorganisms.
14. Clean the wound with the cleaning solution and gauze. Gauze may be held with the forceps, or swabs may be used. Be sure to cleanse from the area least contaminated to the area more contaminated, and use a new swab for each stroke. If there is a drain present, cleanse this area last.	Always clean from the center of the wound to the outer area to avoid contamination of the wound by pathogens present on surrounding skin surfaces.
15. Ifadrainispresent, apply precut dressing around the drain. Apply a thick second layer of gauze over the drain.	To absorb exudate and isolate drainage from the wound.
16. Applysteriledressingoverwound.Then Cover with the surgi pad.	Provides protection of the wound.

17. Securethedressingwitheithertapeortheties from the Montgomery straps. Tape should be placed at the edges of the dressing so that the edges cannot be lifted to expose the wound. Paper tape should be used on patients with thin Fragile skin and patients who have sensitive skin.	Montgomery straps are used when a wound needs to have frequent dressing changes to prevent skin breakdown.
18. Remove gloves, dispose, and wash hands.	Prevents transfer of pathogens.
19. Reassess patient following dressing change to determine status and comfort level.	Determines effect of dressing change; alerts to any patient needs.
20. Document all assessment findings and Actions taken.	Records information for evaluation Of progress of wound healing.

Removing Sutures from Wound

Definition of wound suturing:

Wound suturing is a procedure performed to close the wound from an accident or surgical wound. It is stitch or row of stitches holding together the edges of a surgical incision or wound.

Suture removal

It is a procedure to take sutures out of your skin. Suture removal helps prevent scarring and tissue damage.

Purposes of wound suturing:

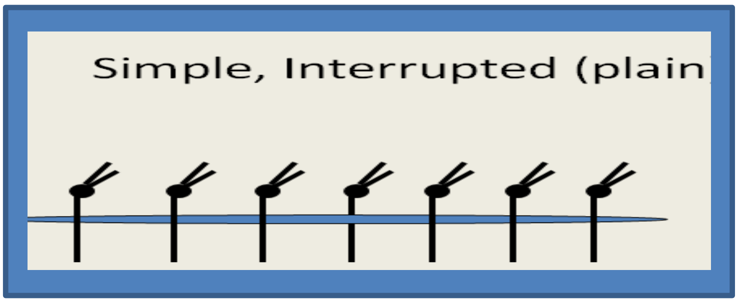
- To stop bleeding, reduce pain and infection,
- To repair the cutaneous wound,
- To minimize scarring, and maximize wound healing.

A-Types of Sutures (according to material):

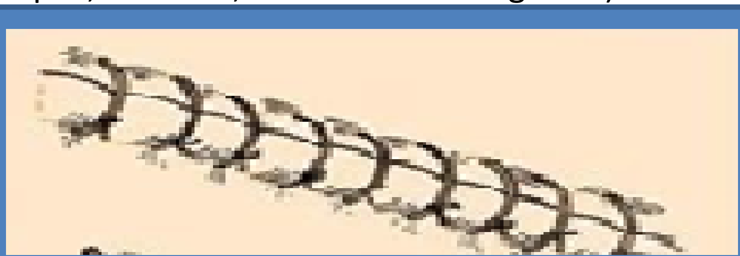
1. Absorbable (eg; catgut, chromic gut) or non-absorbable (eg; Nylon, Prolene, Silk).
2. Monofilament (eg; : nylon, monocryl, prolene), or multifilament (braided)eg; vicryl, silk.
3. Natural (eg; Catgut ,Silk ,and Chromic catgut) ,or synthetic (eg; Nylon Vicryl, Monocryl, and Prolene),

B- Types of Sutures according to structure techniques:

1- Interrupted (plain/ simple as figure1 ,or subcutaneous as figure 2) suture.



2- Continuous (plain/ simple ,Mattress, Subcuticular as figure 6) suture.



Simple continuous

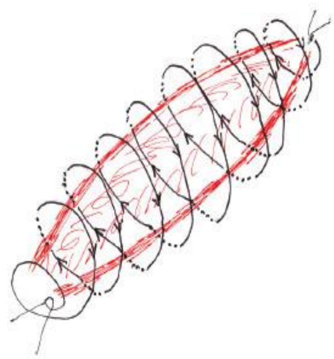


Figure 5. Cutaneous continuous suture

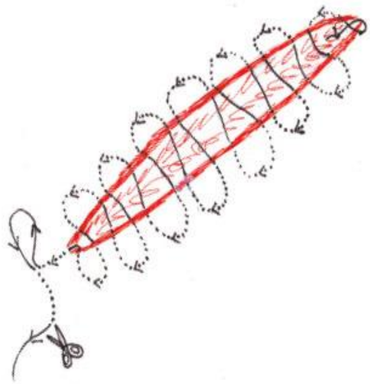
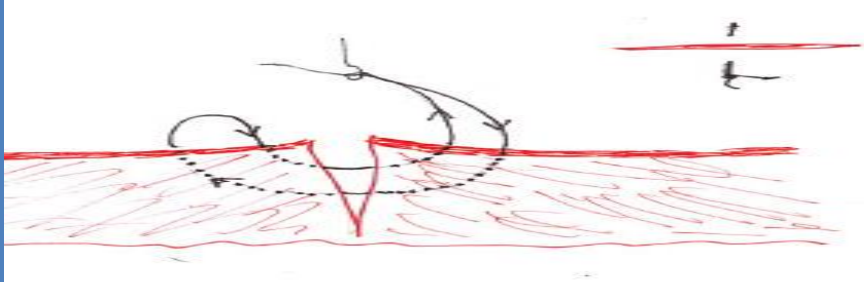


Figure 6. Subcuticular suture

3- **Vertical mattress suture:** Good for everting wound edges (neck, forehead creases, concave surfaces)



3. Vertical mattress suture

Suture Sizes:

the smaller the number the larger the thread. For example: **A 1-0** is larger than**a 6-0** suture. Sizes 3 to 12-0 (numbers alone indicate progressively larger sutures, whereas numbers followed by 0 indicate progressively smaller).

Smaller ←-----------------------------------→Larger

….."3-0"…"2-0"…"1-0"…"0"…"1"…"2"…"3"…..

Types, characteristics and uses of common suture materials		
Type	**Characteristics**	**Use**
Nylon	Non-absorbable, monofilament	Any type of cutaneous suture
Polypropylene	Non-absorbable, monofilament	Any type of cutaneous suture
Silk	Non-absorbable, polyfilament	Any type of cutaneous suture
Polyglactone	Absorbable, polyfilament	Subcutaneous sutures
Poliglecaprone	Absorbable, monofilament	Subcuticular sutures

Principles of Suture Removal:

1-Use sterile forceps or hemostats and proper suture scissors (sharp point down).

2- Cut the suture at skin level.

3-Pull the suture gently and smoothly.

4- Do not pull a contaminated suture through the suture route.

5-Support the suture at the suture exit point.

6-The location of the knots should be noted so that they can be removed first.

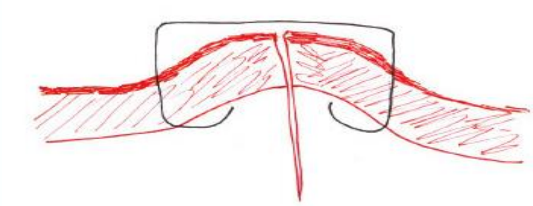
7- Pull the suture up and towards the wound at a 45-degree angle.

Timing of suture removal	
Wound location	**Timing of removal (days)**
Face	3–5
Eyebrow, Nose, Lip	3-5
Scalp	7–10
Arms	7–10
Chest and abdomen	8-10
Torso	10–14
Legs	10–14
Dorsum of hands or feet	10–14
Palms or soles	14–21
Patient’s condition can delay wound healing: (Corticosteroid use, diabetes mellitus)	14 to 21 days

Staples:

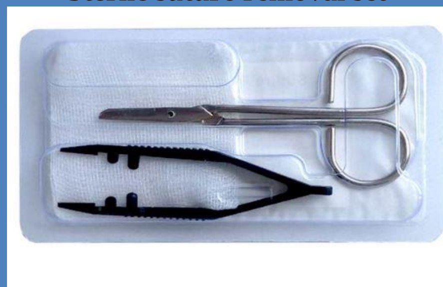
Commercially available single-use stapling devices , it may be an alternative to simple interrupted sutures where rapid wound closure is needed. Staples are made of stainless steel, which does not react with tissues. Application involves less tissue trauma than regular sutures, thereby theoretically reducing the risk of wound infection. Removal of staples special tool was required

Single-use skin stapler



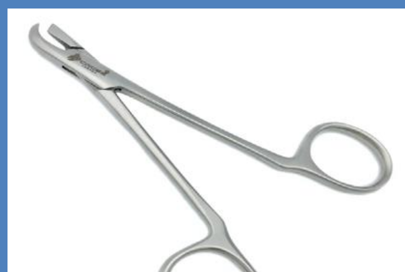
Wound edge eversion by skin stapler

Sterile suture removal set



Equipment:

- Sterile suture removal set (forceps and scissors)
- Disposable water proof biohazard bag
- Sterile antiseptic swabs
- Gauze pads
- Steri-Strips or butterfly adhesive strips
- Clean gloves(sterile gloves optional)

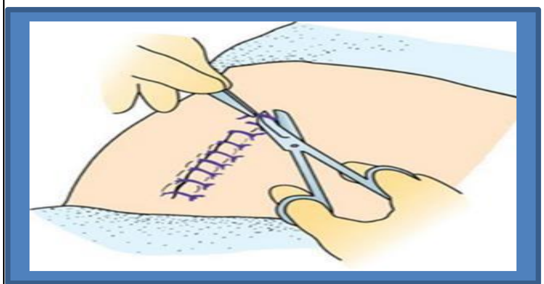


Clip suture removal

Forceps with suture and thread

Procedure

Action	Rationales
1.Confirm the physician's order.	
2. Assess the patient's pain level and medicate with analgesic 30 minutes before procedure if the medication is to be given by oral or IM.	Allows time for medication to be absorbed to increase the analgesic effect.
3-Prepare necessary equipment	
3.Explaintheproceduretothepatient.	Helps to decrease the patient's anxiety and increase the patient's cooperation.
4- Keep the patient privacy.	
5.Placeawaterproofpadonthebed.Assist the patient onto the pad. Then assist the patient into comfort position.	Positioning of the patient and placement of a waterproof pad will decrease contamination of bed linen.
6. Wash hands and done disposable gloves.	Prevents transmission of microorganisms.
7. remove the old dressing and clean the incision with antiseptic swabs.. Start at sides next to incision and then wipe across suture line using new antiseptic swab for each swipe.	- to decrease risk of infection.  - The suture line is usually cleaned with antiseptic solution before and after suture removal.as prophylactic measure to prevent infection.

8.Assessthewound’s site for :appearance, color ,suture site before removal and presence of pus or drainage(indicate presence of infection).	Provides assessment of the status of the wound to confirm healing of wound or incision before removal of suture.
9-Don sterile gloves.	To prevent microorganisms
10-Opensterilepackagesofequipment needed for suture/staple removal: a. Open sterile suture removal kit b. Open sterile antiseptic swabs and place on inside surface of kit.	Ensures an organized procedure. Avoids contamination of the antiseptic swabs.
11. Removeinterruptedsutures: Place gauze few inches from suture line. Hold scissors in dominant hand and forceps (clamp) in non-dominant hand.	Gauze serves as receptacle for removed sutures .Placement of Scissors and forceps allows for efficient suture removal.
12. Grasp the suture at the knot with a forceps.	
13- Gently pull up knot while slipping tip of scissors under knot of suture near skin	To Release suture.  Removal of continuous suture. Nurse cuts suture as close to skin as possible, away From knot.
14-Put the curved tip of the suture scissors under the suture as close to the skin as possible or directly under the knot.	
15.with the forceps, gentle pull the suture out in one a piece. Be sure that all suture thread is removed.	-Smoothly removes suture without Additional tension on suture line.
16-Discard the suture onto a piece of sterile gauze. being careful not to contaminate the forceps tip sometimes the suture sticks to the forceps and need to be removed by wiping the tip on the sterile gauze.	-suture threads (materials) left beneath the skin act as a foreign body and causes inflammation.

17-Continue to remove altenate sutures eg, third, fifth, seventh, ..etc.	-Alternate sutures are removed first so that remaining suture keep the skin edges in close approximation and prevent dehiscence from becoming large.
18- If no dehiscence occurs, removed the remaning sutures , if dehiscence occurs report to the physician. a- If a little wound dehiscence occure, apply a sterile butterfly tape over the gap and follow the following; - Attach the tape to the one side of incision. - Press the wound edges together. - Attach the tape to the other side of the incision. b- If large dehiscence occurs, cover the wound with sterile gauze and report the problem immediately to the physician.	Repeat until you have removed every other suture.  -the butterfly tap hold the wound edges as close together as possible and promotes healing. -to prevent exposure of the wound to the contaminated atmosphere.
19 -clean the incision again with antiseptic swabs.	-to prevent infection as prophylactic measures.
20. Apply Steri-Strips if any separation greater than two stitches or two staples in width is apparent, to maintain contact Between wound edges.	Steri-Strips support wound by distributing tension across wound and eliminate scarring from closure techniques.
21-Apply sterile dressing as Steri-Strips over incision	 - Application of a sterile dressing protects the wound from microorganisms and friction with clothes.

22. Remove sterile gloves and wash hands.	
23-Instruct the patient about follow up of wound care specially after suture removal and keep wound incision dry and healing well patient.	-to prevent complications and to help early detection of any problem.
24. Document the sutures removal and relevant assessment of wound healing.	Records information for evaluation.

Intravenous Therapy (IV Therapy)

Definition:

It is an effective and efficient method of supplying fluid directly into intravenous fluid of compartment producing rapid effect with availability of injecting large volume fluid more than other method of administration

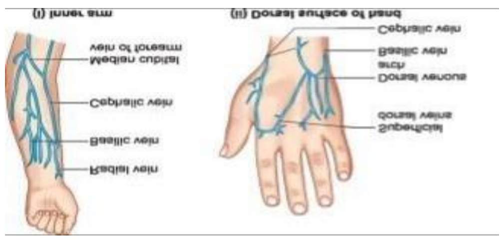
Intravenous therapy consists of medications, fluids, blood& plasma given by I.V route. Or for diagnostic tests.

Indications of IV therapy:

- Provide fluids electrolyte maintains, replacement.
- Administer medications for nutritional feeding.
- Administer blood & blood products
- Keep a vein open for quick access.
- Maintain urine output.

Vein puncture sites: -

- Basilic & cephalic veins in the antecubital space of forearm are most common sites these are large and near the skin surface.
- Hand veins: Cephalic, Basilic, Dorsal Venous network and Dorsal Metacarpal Vein.
- Median cubital vein.
- Ulnar Vein.
- Radial Vein.



Vein character to be used: -

- The vein should be large, visible, straight, superficial, palpated, strong in non-dominant arm, if possible.
- Avoid areas that are painful to palpation.
- Choose a site that will not interfere with planned procedures or the patient's activities of daily living

What Veins the Nurse Must Avoid?

- Avoid vessels in an extremity with compromised circulation such as in cases of mastectomy, dialysis graft, or paralysis.
- Avoid sites distal to previous venipuncture sites, hardened cord like vein, infiltrated sites, and bruised areas, (such areas cause infiltration of new IV lines

and excessive vessel damage.)

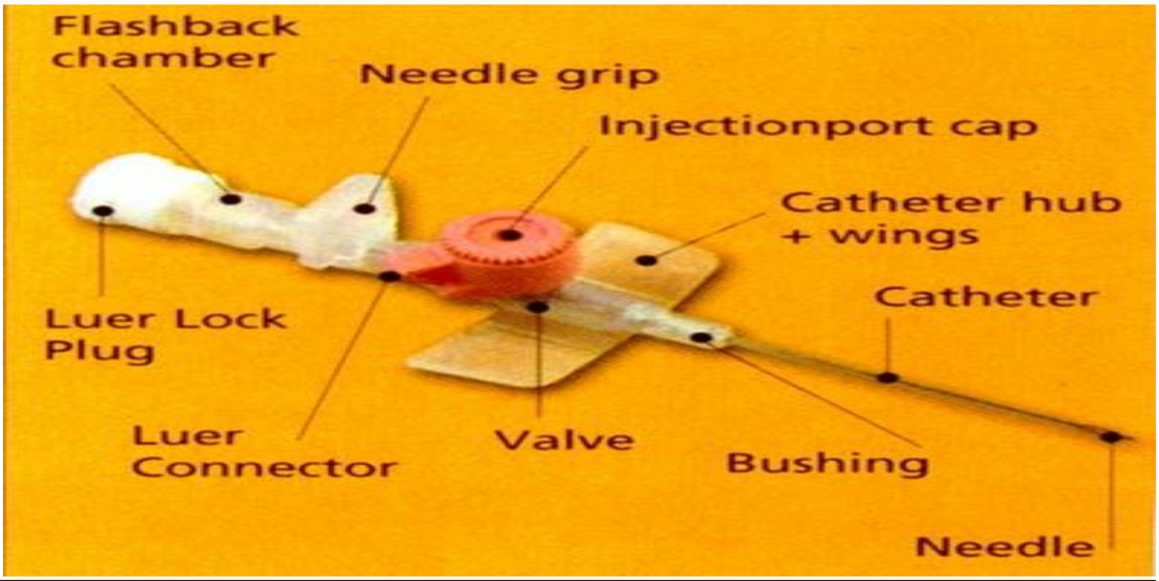
- Thrombosis / fibroid
- Inflamed
- Thin / fragile
- Mobile
- Near bony prominences
- Areas or sites of infection, oedema or phlebitis
- Have undergone multiple previous punctures

Equipment: -

- Intravenous infusion tray, including (tourniquet, alcohol swab, adhesive tape)

- Extension tubing and stand.
- IV catheter (needle/cannula)
- The prescribed fluid, medication stopped being given.
- Saline syringe
- Clean gloves.
- Sterile gauze or transparent dressing

Parts of cannula:

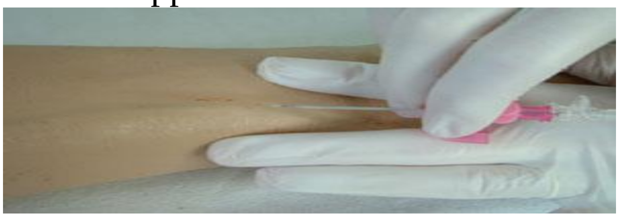


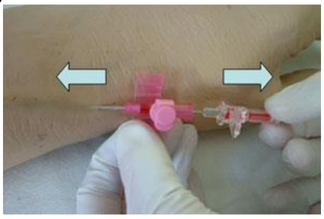
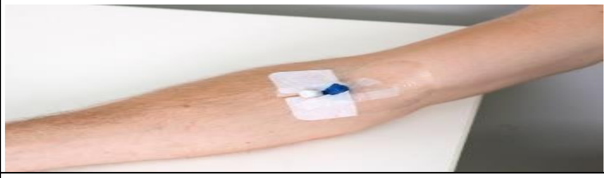
Different sizes of adult devices

Color code	Gauge and length	Flow rate (mL/mm)
Orange	14G x 1 3/4"	270
Grey	16G x 1 3/4"	180
White	17G x 1 3/4"	125
Green	18G x 1 3/4"	80
Pink	20G x 1 1/4"	54
Blue	22G x 1"	33
Yellow	24G x 1 3/4"	20
Violet	26G x 1 5/8"	19

Procedure:

	Steps	Rational
1	Review physician's orders and medications record.	To ensure safe and correct administration of IV therapy.
2	Assess for clinical factor that will respondtooraffectedbyIVfluidsuchasperipheraledema/dryskin/poorskinturgor.	To provide baseline to determine effect on patient fluid and electrolyte balance
3	**Prepare equipment.** - Organize equipment - Prepare IV infusion tubing and solution by check iv solution using five right of medication administration and check solution for color, clarity and expired date. - Cut pieces of tap and place on edge of clean over -bed table.	To reduce risk for error.  Tc tin eff
4	**Prepare environment** - By adjust lighting as necessary. - Close room, door and curtains.	To keep privacy.
5	Hand washing and wear gloves.	To reduce transmission of infection.
6	Verify patient identity by use at least two patient identification.	
7	Introduce yourself to the patient & explain procedure	- To improve patient safety. - To relieve PT anxiety &gain his Cooperation
8	Assist patient to comfortable position.	To promote comfort and relaxation of patient.
9	Put gloves, ask the patient to open and close the fist two times.	To destined the vein
10	Inspect both of patient's arms, palpating and visualizing the vein.	To promotes patient movement& Comfort and adequate vein for infusion.
11	Apply tourniquet around arm 10 to 15cm above the selected site (should be tight enough to impede venous return but not occlude arterial flow).	To facilitates observation and puncture of distended veins.

12	**Select venipuncture site.** • Palpate the vein by pressing downward and noting the resistant, soft, bouncy feeling as pressure is released. • Promote venous distention by instructing the patient to open and close the fist several times and hold it closed until needle is in the vein. • Lowering the patient arm and rubbing or tapping the patient arm against the direction of circulation.	 
13	Cleanse site 2-4 inches in diameter with alcohol swab, start at the center, and work in circular motion allow the area to dry**.**	To remove microorganisms from puncture site.
14	•Remove the needle cover and holding IV cannula bevel up. •Stretch skin taut and stabilize vein with a non-dominant hand. •Insert cannula at 30-degree angle and reduce angle when flashback occurs. lower angle to 10 degrees and enter vein wall. Slight resistance.	To prevent vein from moving during Procedure. Allows needle to enter smoothly through skin and approach vein wall.  Reduces the risk of going completely through vein and ensures placement Within vein
15	Watching blood return through cannula backflow chamber.	Permit venous flow, reduce back flow of blood. 
16	Stabilize the cannula with one hand and release tourniquet.	To confirm placement within vein. 

17	Holding guide needle in place, gently thread plastic catheter off needle and into vein.	To decreases potential for vein rupture.
18	• Apply gentle pressure over catheter in vein, remove guide needle, and attach sterile connection end of primed tubing into catheter hub. • Advance the cannula while simultaneously withdrawing the needle.	To prevents needle going through catheter &vein ,and prevents catheter from being dislodged by slightly movement.  
19	Ask the patient to open and relax his hand.	
20	Following agency policy, secure and dress site with tape, and dressings.	To maintain catheter in place, prompts infusion and avoids clotting of blood in catheter. To prevents unintentional dislodging of catheter from vein and prevents infection. 
21	**Flush The Cannula** • Now flush the cannula with saline. Some guidelines would suggest using an extension when introducing medication via this port.	This allows to check the patency of the cannula. 
22	Label site and tubing according to Agency policy.	Serves as reminder to change site and tubing.
23	Open the valve and connect IV-line therapy, then adjust fluid flow rate according to accurate drop rate calculations.	To provide fluid therapy according to Medical order.
24	• Return and discard equipment. • Remove gloves. • Wash hands.	

25	• ***Documentations and report:*** - Date & time - Site and size of cannula - Any problems encountered - Type of fluid, flow rate, size of needle. • Immediately report for any abnormal reactions.	

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Chest-physiotherapy

- **Definition:**

A chest physiotherapy treatment is a group of airway clearance techniques that uses to mobilize the lung and manually percuss the chest wall to help clear mucus build up in the lungs and improve lung function and help to patient breathe better.

- **Purpose:**

1. To reduce airway obstruction by improving the clearance of secretions.
2. To reduce the severity of the infection by clearing infected material.
3. To maintain optimal respiratory function and exercise tolerance.

- **Indication of chest physiotherapy:**

Secretions accumulate in the airways of patients with:

- Lung diseases such as Bronchitis, Asthma and Pneumonia.
- Bronchiectasis and Atelectasis.
- Mucous plugs and Lobular collapse, collapse occurs when secretions accumulate in the airways.
- Surgical patients in the post-operative period sometimes have excess secretions because of anesthesia and ineffective coughing due to incision pain.

- **Contraindications of chest physiotherapy:**
  - Increased ICP" intracranial pressure".
  - Unstable head or neck injury.
  - Active hemorrhage with hemodynamic instability or hemoptysis.
  - Recent spinal injury.
  - Empyema.
  - Bronco- pleural fistula.
  - Rib fracture.
  - Fail chest.
  - Vertebral fractures or osteoporosis.

- **Technique use in chest physiotherapy:**

1- Chest percussion.
2- Chest vibration.
3- Postural drainage.
4- Deep breathing exercise.

1. Chest percussion

➢ Definition: -

Rhythmically striking the chest wall with cupped hands. The purpose of percussion is to break secretion on the wall of the lung and improve ventilation. It is done with the middle finger of one hand tapping on the middle finger of the other hand using wrist action. The non-striking hand(finger) is placed body over the tissue.

➢ Procedure: -

<table>
<tr><th>Procedure</th></tr>
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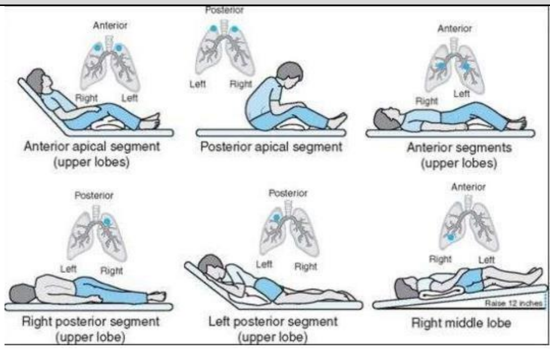
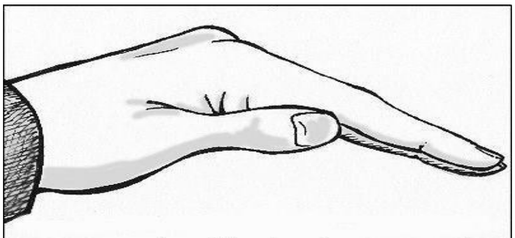
- Lie in a postural draining position, i.e., with your chest elevated above your head, so affected area is elevated. Lie on the left side if the right side is to be percussed, or, lie on the right side if the left side is to be percussed.



- Place a towel over the area to be percussed.
- Hand should be cupped, with the tip of the thumb at the side of the middle joint of the index finger. Fingers should be straight and held closed.


Correct Hand Position for Chest Percussion

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- With hand in this cupped position, forcefully tap the rib cage over the affected area, using a smooth rhythm and keeping the arms and shoulders relaxed.

- Percussfor1-2minutes.Alwaysavoidpercussingoverthespinal column and the kidneys.
- Use suction or assisted cough, following each position.

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2- Chest vibration

- **Definition**: -

Vibration involves the application of a fine tremorous action (manually performed by pressing in the direction that the ribs and soft tissue of the chest move during expiration) over the draining area.

- **Procedure: -**

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<tr><th>Procedure</th></tr>
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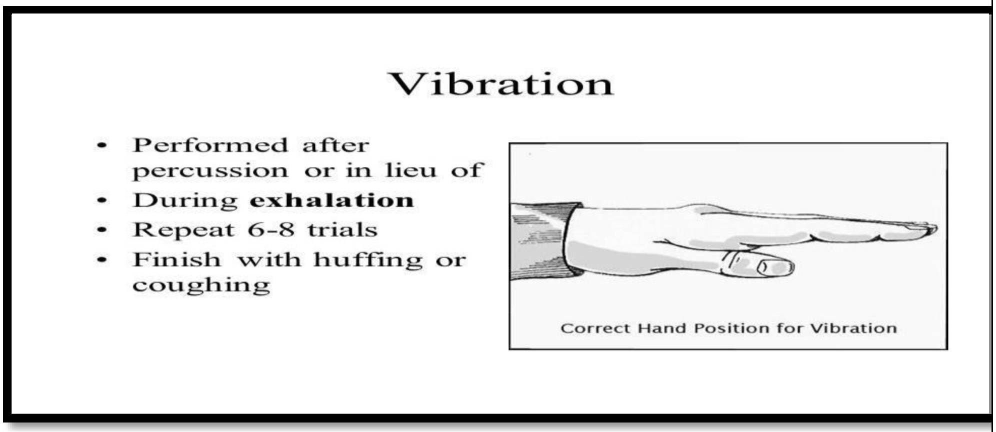
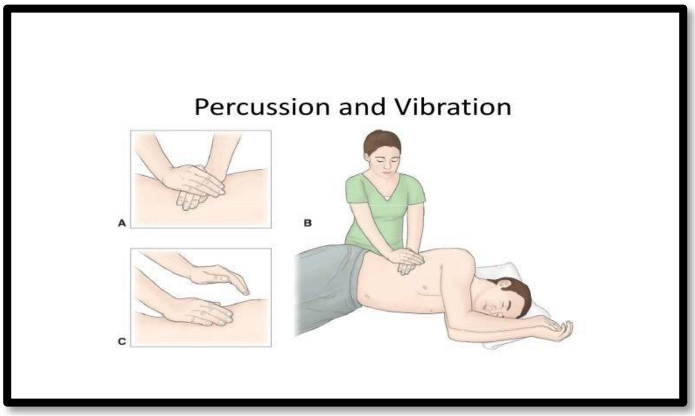
It is performed as the patient breathes deeply.
When done manually, the person performing the vibration places his or her hands against the patient's chest and creates vibrations by quickly contracting and relaxing arm and shoulder muscles while the patient exhales. The procedure is repeated several times each day for about five exhalations.





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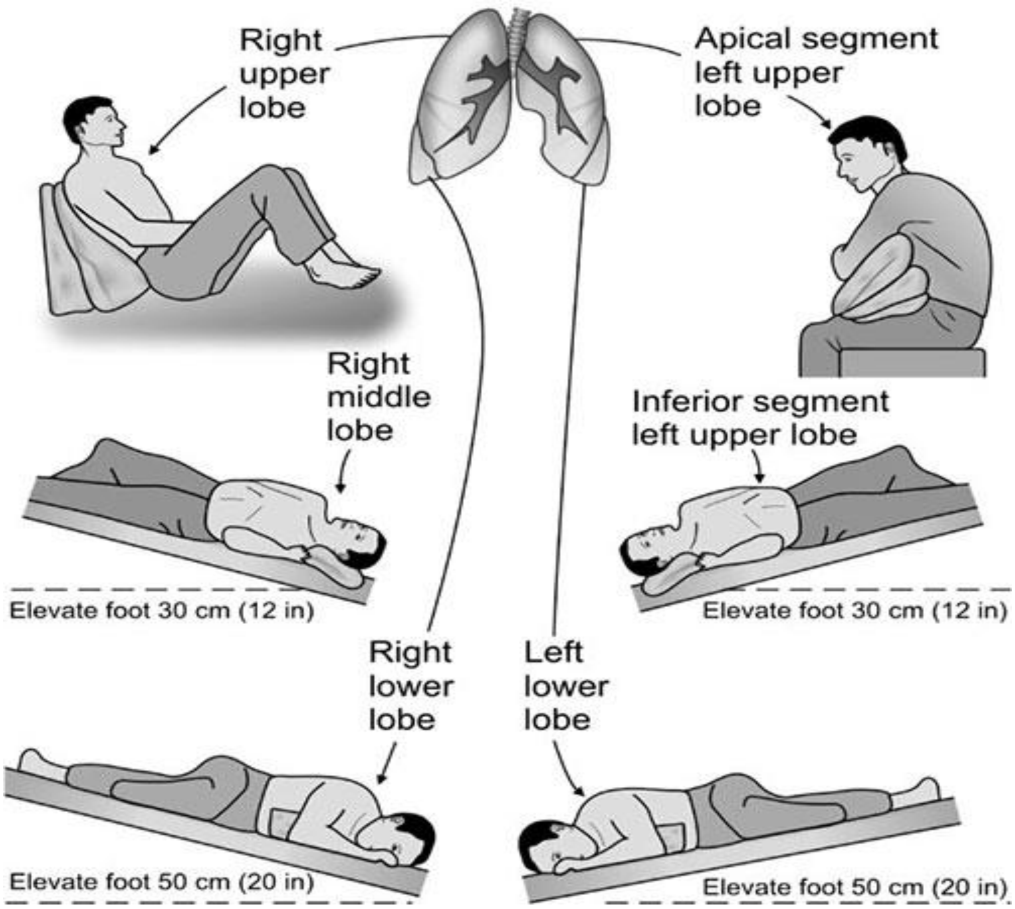
3- Postural drainage

- **Definition:-**

  Postural Drainage involves a patient assuming various positions to facilitate the flow of secretions from various parts of the lung into the bronchi, trachea, and throat, so that they can be cleared and expelled from the lungs more easily.

- **Postural drainage positions (Positioning and turning from side to side):**

The right lung is divided into three lobes (right upper lobe, right middle lobe and right lower lobe) while the left lung has only two lobes (left upper lobe and lower lobe). The figure below shows the individual segments which make up each lobe.



Procedure:-

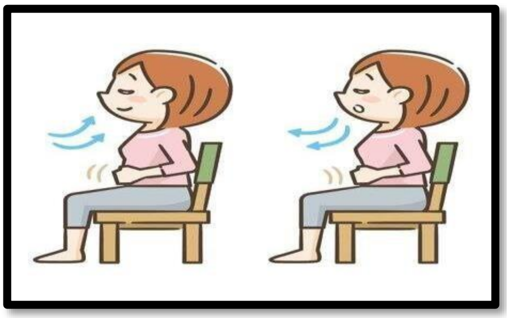
Step	Rational
Explain procedure to the patient.	To gain his cooperation.
Cover the Patient with a sheet and Expose her chest.	To provide privacy.
Wash hand and Wearing PPE	To reduce transmission of micro-organisms and protect yourself from infection.
Close windows and doors in the Examining room.	To avoid air streams and to maintain privacy.
Eliminate noise as possible as you can	-To ensure hearing.
The chest should be osculated before and after the procedure.	To determine areas needing drainage and the effectiveness of the treatment.
If prescribed, bronchodilators, water or saline may be nebulized and inhaled Before posture drainage.	To decrease thickness of mucus and sputum.
Posturaldrainageisusuallydone2to4 times daily before meals and bedtime.	To avoid nausea, vomiting and aspiration.
Patient is instructed to remain in each positionfor5to10minandtobreathe in slowly through his nose and then breathe out slowly through pursed lips.	
When the patients change positions, he is instructed cough to remove secretions. Cough twice during each exhalation while contracting the abdomen.	
Following the procedure, record the amount, color, viscosity and character of the ejected sputum.	
Encourage mouth care after postural drainage.	
Hand Washing	To prevent Infection
Clean and Remove equipment.	
Document findings related to Postural drainage(amount, color, viscosity and character of the ejected sputum).	-Proper documentation saves time and Facilitate early detection of any deviations

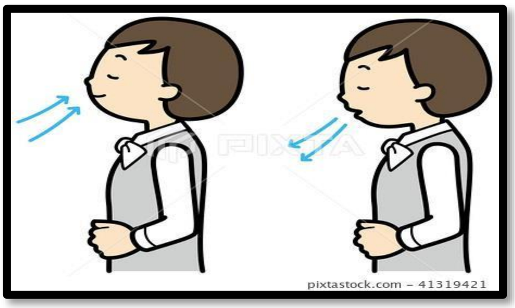
4- Deep breathing exercise

Definition:

Take a slow deep breath in through your nose the maximum amount as you'll be able to. Attempt to keep your chest and shoulders relaxed take a breath gently and relaxed, It is easy and smooth movement to mobilize respiratory muscles.

Procedure:
▪ Sit or stand, pull your elbows back firmly, and inhale deeply. ▪ Hold your breath for 5 counts. ▪ Exhale slowly and completely.





Nasogastric Tube Insertion Procedure

Definition:

Nasogastric tubes (NG) are tubes inserted through the nares to pass through the posterior oropharynx, down the esophagus, and into the stomach. Nasogastric tubes be used to administer nutrition or medication to patients who cannot tolerate oral intake. Additionally, it can be used for decompression of the stomach in intestinal obstruction or ileus.

Indications for NG tube:

- Gastric decompression, including maintenance of a decompressed state after endotracheal intubation or to relieve distention due to obstruction, ileus, or atony).
- Relief of symptoms and bowel rest in small-bowel obstruction
- Aspiration of gastric content from recent ingestion of toxic material
- Administration of medication
- Feeding
- Bowel irrigation
- NGT can be kept following corrosive ingestion for the development of a tract in the esophagus that subsequently can be used for balloon dilatation
- Removal of blood and clots in patients with GI bleeding.
- Removal of ingested toxins (rare)
- To give antidotes such as activated charcoal
- To provide feeding of nutrients into stomach or feeding directly into small intestine with a long, thin, flexible enteral feeding tube

Diagnostic indications include:

- To give oral radiopaque contrast agents.
- To obtain a sample of gastric contents to assess bleeding, volume, or acid content.

Contraindications for NG tube:

- Absolute contraindications:
  - Mid face trauma
  - Recent nasal surgery
- Relative contraindications:
  - Coagulation abnormalities
  - Recent alkaline ingestion (due to risk of esophageal rupture)
  - Esophageal varices (untreated or recently banded/cauterized)

- Esophageal strictures

Complications of NG tube insertion:

- Gagging or vomiting
- Nasopharyngeal trauma with or without hemorrhage
- Sinusitis and sore throat
- Pulmonary aspiration
- Traumatic esophageal or gastric hemorrhage or perforation

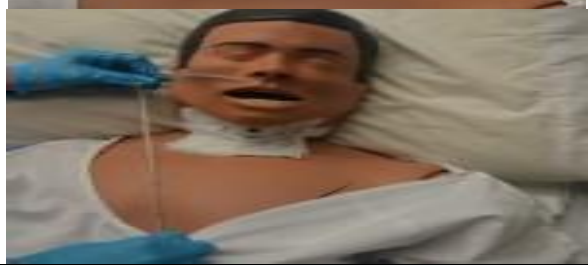
Equipment

Gather the relevant pieces of **equipment** and place into a tray:

- Nasogastric tube (fine bore)
- Disposable gloves
- Lubricant and gauze: to lubricate the tip of the NG tube.
- Disposable bowl: to be used in the event of vomiting.
- Paper towels: to allow the patient to wipe around the mouth if needed.
- Large syringe: to obtain an aspirate from the Ng tube.
- PH testing strips: to assess the pH of the aspirate.
- Dressing: to secure the NG tube.
- A glass of water for the patient (if swallow is deemed safe).

Procedure

Steps	Rational
1- Check doctor's orders for type of NG tube to be placedand reason for placement (to determine whether the NG tube is to be attached to suction or a drainage bag.).	- Check appropriate orders relevant to patient safety.
2- Explain the procedure to the patient. • Agree on a signal the patient can use if they wish you to pause during the procedure.	This procedure can be anxiety-provoking and uncomfortable for many patients. Providing a means for the patient to communicate discomfort and a desire to pause during the Procedure helps alleviate anxiety.
3- Perform hand hygiene and gather supplies.	- This prevents the transmission of microorganisms. 
4- Patient's assessment: • Visually inspect condition of patient's nasal and oral cavities. • Assess for the best nostril before beginning: by occluding one side and asking the patient to breath from the other. • Ask the patient about previous injuries or history of a deviated septum. • Palpate patient's abdomen for distension, pain, and/or rigidity. Auscultate for bowel sounds. • Assess patient's level of consciousness and understanding of procedure.	- Check for signs of infection or skin breakdown. - If either nostrils equally suitable, select the nostril closest to the suction. - Assess foremost patent nostril.  - Document assessment findings and determine appropriateness of NG tube insertion related to reason for insertion and patient's physical assessment. - Patient must be able to follow instructions related to NG insertion to allow for passage of tube through nasal and gastrointestinal tracts.

5- Positioning: • Position patient sitting up at 45 to 90 degrees (unless contraindicated by the patient's condition), with a pillow under the head and shoulders. • Raise bed to a comfortable working height. • Stand on patient's right side if right-handed and the left side if left-handed.	- This allows the NG tube to pass or easily Through then asopharynxandin to the stomach. - To use the dominant hand to insert the tube.
6- Place a towel on the patient's chest and provide facial tissues and an emesis basin.	Nasal and oral secretions may be evident during the procedure.
7- Apply clean non-sterile gloves.	- Using gloves decreases the transfer of microorganisms. 
8- Measure distance of the tube from: ▪ The tip of the nose to the earlobe ▪ Then, from the earlobe to the xiphoid process and then mark the tube at this point.	This determines the appropriate length of NG tube to be inserted.  
9- Lubricate NG tube tip according to the agency policy.	Tube may be lubricated internally using tap water or externally using water-soluble lubricating jelly. Agency policy varies and Should be checked. 

10.Curve10to15cmoftheendoftheNG tube around gloved finger and then release it.	Curling the NG tube around the finger helps it conforms to the normal curve of the nasopharynx. 
11. Keep patient's head hyperflexed and breathe through the mouth.	- Hyperflexion closes the trachea and opens the esophagus, which allows the NG tube to pass more easily through then asopharynx and into the stomach.
12. If slight resistance felt as advance along the nasal passage: • Twist the tube slightly, apply downward pressure, and continue trying to advance the tube. • Ask the patient to sip water slowly through a straw unless oral fluids are contraindicated. • If oral fluids are not allowed, ask the patient to try dry swallowing while advancing the tube. • If significant resistance is felt, remove the tube and allow the patient to rest before trying again in the other nostril.	- It is common for the patient to feel discomfort, and this may be expressed with light coughing and gagging. More aggressive coughing and gagging may indicate that the tube has entered the airways, in which case NG tube should be withdrawn.  - If patient continues to gag or cough, check that the tube is not coiled in the back of the mouth, using a tongue blade and a flashlight to check the back of the mouth. If tube is coiled, withdraw the tube until only the tip of the tube is seen in the back of the mouth. Then try advancing the tube again while patient tries to swallow.
13. Continue to advance NG tube until reaching the mark/tape that had placed for measurement.	- This ensures accurate placement.

14. Temporarily anchor the tube to patient's cheek with a piece of tape until you can check for correct placement.	- This prevents displacement of the NG tube while checking placement. 
15. Verify tube placement according to agency policy. Colour-coded pH paper is usually used, as an initial and interim check, to confirm that acidic contents are present. Then an X-ray is taken to confirm placement prior to using NG tube for feeding.	The contents aspirated from the tube should be acidic with a pH<5. If the pH is more than 6, it may indicate the presence of respiratory fluids or small bowel content, and the tube should be removed. 
16. Once the tube placement has been confirmed, mark (with a permanent marker) and record the length of tubing extending from the nose to the outer end of the tube.	This aids in timely recognition and identification of tube displacement or migration.
17. Secure the tube to the patient's gown, allowing enough tube length for comfortable head movement.	This keeps the NG tube in place.
18. Document the procedure according to agency policy and report any Unexpected findings.	Timely and accurate documentation promotes patient safety.

Special considerations with NG tubes:

- Always assess correct placement of the NG tube prior to infusing any fluids or tube feeds as per agency policy. Check location of external markings on the tube and colour of the PH of fluid aspirated from the tube. Routine evaluation will ensure the correct placement of the tube and reduce the risk of aspiration. Do not instill air to test location of tube.

- Do not give the patient anything to eat or drink without knowing that the patient has passed a swallowing assessment.

- If changing the gown or repositioning the patient, take care not to pull on the NG tube. Remember to unpin the tube from the gown and repine the tube.

VIDEO

https://www.youtube.com/watch?v=z5PZXA1zqA4